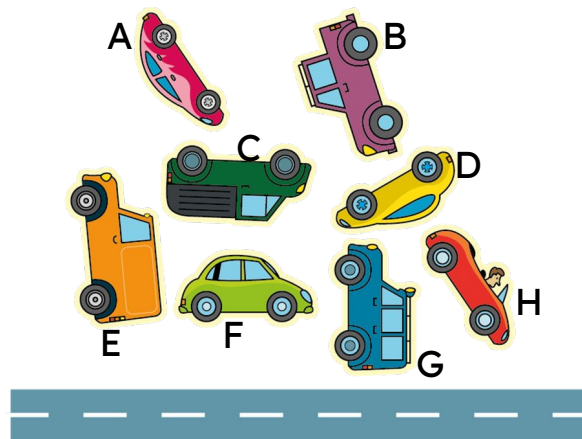
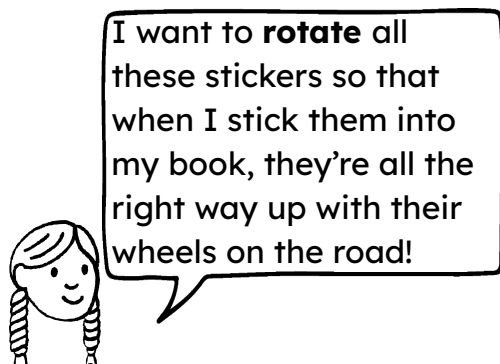




Introduction to rotations

Task A: Identifying rotations in the real world



- 1) Which vehicles will be facing left, and which will be facing right?
- Left: A, B, C, E, G
Right: D, F, H

2) Sketch your own vehicles in different orientations.

Can you draw two vehicles that face the same way, even though you have to **rotate** them differently?

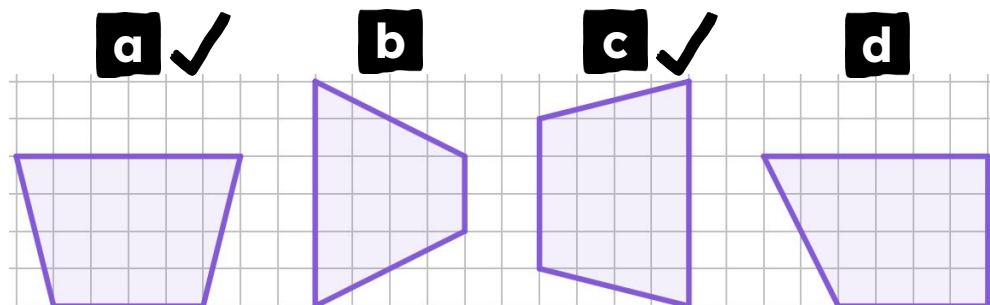
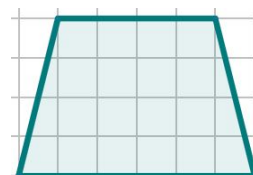
For example:



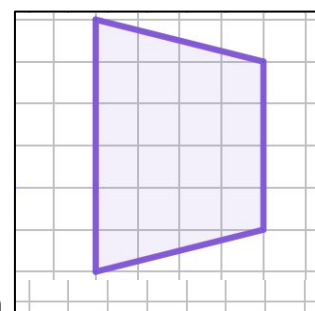
Both these cars face right when **rotated**.

Task B: Understanding properties of rotations

- 1) Select all of the images which are **rotations** of this trapezium.



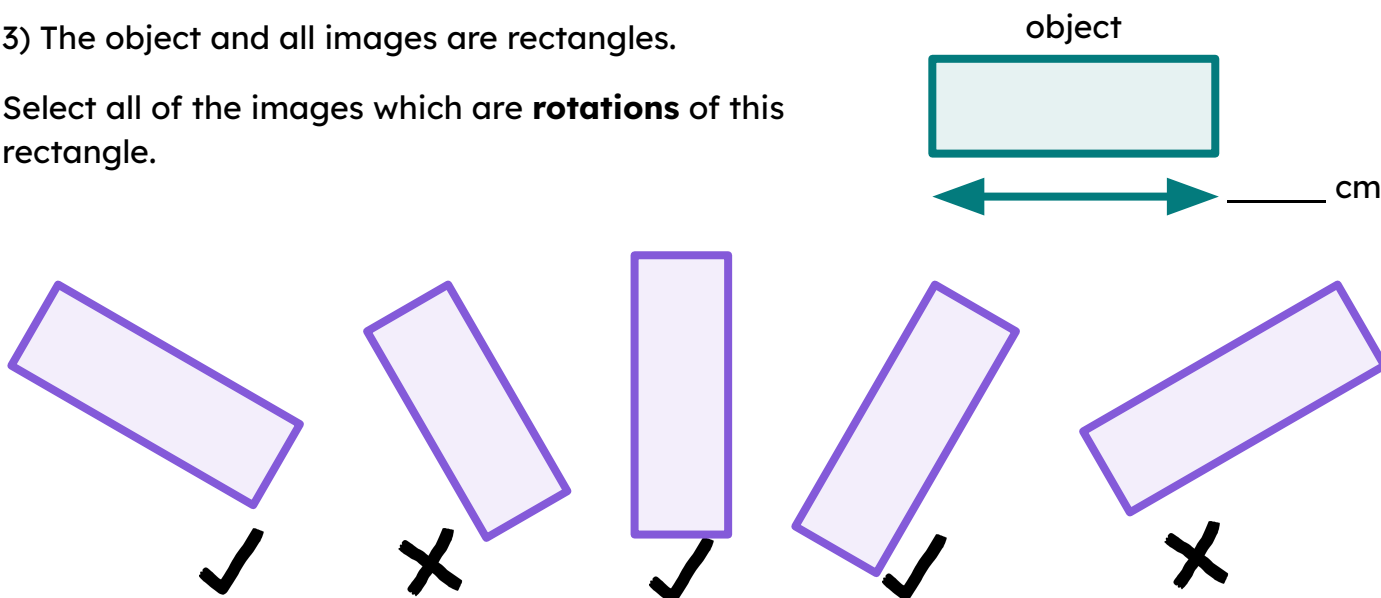
- 2) Using a ruler, draw a different correct **rotation** of the trapezium in the space given.



Draw your own

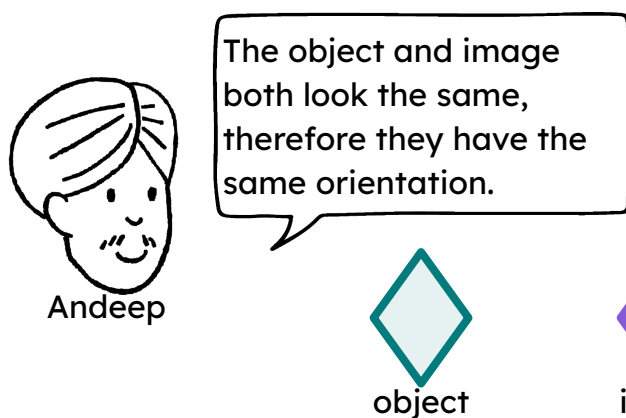
3) The object and all images are rectangles.

Select all of the images which are **rotations** of this rectangle.



4) The object has been **rotated** to become the image.

Explain why Andeep might be incorrect.



*Object and image only have the same orientation if the object has been **rotated** through one full turn, or multiple full turns.*

*If it has been **rotated** by less than that, then the image may look invariant, but still has a different orientation.*

5) Complete the table with ✓ to show the properties of **rotation**.

	Always Invariant	Usually Changes
Lengths of Sides	✓	
Size of Angles	✓	
Position		✓
Orientation		✓

